

PH 605: Superconductivity

Course Code	PH 605
Course Name	Superconductivity
L-T-P-C	3 0-0-3
Prerequisites	Quantum Mechanics-1(PH 513), Condensed Matter Physics (PH 523)
Intended for	PhD/PG/UG
Distribution	Elective
Approval	27th BoA (Nov 2018); 19th Senate (Feb 2019)

Course Contents

- **Introduction**
- Historical perspective, Resistivity, Specific heat, Thermal conductivity, Magnetic Susceptibility and Hall Effect of normal metal; Zero resistance, persistent Current, Meissner effect, London-London equations, Penetration depth and critical field. [9 Lectures]
- **Phenomenon logical theory of Superconductivity**
- Free energy, First order and second order transition, specific heat, thermal conductivity, Superconducting order parameter, Ginzberg-Landau equations and its predictions, Coherence length, Type-I and Type-II superconductors, The vortex lattice. Phase coherence, Flux quantization. [9 Lectures]
- **Microscopic Theory of Superconductivity**
- Isotope effect and its significance, The Cooper problem, Formation of Cooper pairs, BCS wave function, Existence of energy gap, Finite temperature properties of BCS ground state [9 Lectures]
- **Tunneling and energy gap**
- Tunneling phenomenon, DC Josephson Effect AC Josephson, Effect, Inverse AC Josephson Effect and Shapiro jump , supercomputing quantum interference device (SQUID).[7 Lectures]
- **Unconventional Superconductors**
- Alternate pairing mechanisms (e.g, spin-triplet, d-wave etc.), Symmetry of the gap function, Experimental methods for probing Nodal structure, Parity, spin state, Lattice symmetry and internal structure, Heavy Fermion, High temperature superconductivity, Cuprates, and Fe based Superconductors. [8 Lectures]

Text Books

1. J.B. Ketterson and S.N, Song, **Superconductivity**, Cambridge University Press, 1999.

References

1. M. Tinkham, **Introduction to Superconductivity**, McGraw-Hill, 1996.
2. Grenel Goll, **Unconventional Superconductors**, Springer-Verlag, 2006.
3. Charles p. Poole Jr., H.A. Farach, R.J. Creswick, R. Prozorov, **Superconductivity**, Elsevier, 2007.
4. James F. Annett, **Superconductivity, Superfluids and Condensates**, Oxford University Press, 2004.